Exchangeability and the Dirichlet Limit of a Multicolor Pólya Urn

Route-A source-local certificate HCS-C263

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Abstract

We close the positive-reinforcement classical multicolor Pólya urn from ordered histories to its count law and normalized-mass martingale. The proof is exact for every number of colors and every active positive initial mass. This source probability theorem uses no target arithmetic or spectral data.

1 Frozen urn

Let $a_i > 0$ for $1 \leq i \leq K$, let $c > 0$, and put $\alpha_i = a_i/c$ and $A = \sum_i \alpha_i$. At each integer time a color is drawn proportionally to its current mass, replaced, and receives additional mass c. Write $N_i(n)$ for its count among the first n draws and use the rising factorial $(x)^{\overline{r}} = x(x+1)\cdots(x+r-1)$.

Theorem 1 (word and count law)**.** *If an ordered word $w = (w_1, \ldots, w_n)$ has multiplicities n_i, then*

$$\mathbb{P}(w) = \frac{\prod_{i=1}^{K} (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \tag{1}$$

Consequently

$$\mathbb{P}\{N(n) = (n_1, \ldots, n_K)\} = \frac{n!}{\prod_i n_i!} \frac{\prod_i (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \tag{2}$$

Proof. Before draw $t+1$, the predictive probability of color i is $(\alpha_i + N_i(t))/(A+t)$. Multiplication along a word groups the numerator factors by color and yields (1), independent of order. There are $n!/\prod_i n_i!$ words with the given counts, proving (2). Summing (2) gives one by the multivariate rising-factorial Vandermonde identity. □

2 Normalized-mass martingale

Define

$$P_i(n) = \frac{\alpha_i + N_i(n)}{A+n}.$$

Conditioning on the first n draws gives

$$\begin{aligned} \mathbb{E}[P_i(n+1) \mid \mathcal{F}_n] &= P_i(n) \frac{\alpha_i + N_i(n) + 1}{A+n+1} + (1 - P_i(n)) \frac{\alpha_i + N_i(n)}{A+n+1} \\ &= P_i(n). \end{aligned}$$

Thus each coordinate is a bounded martingale and the vector converges almost surely. The next revisions identify the limit law and close all moments and boundary faces; this initial version fixes the chronology and normalization.

3 Certificate and Route-A boundary

The exact implementation enumerates ordered words and independently aggregates them by composition. It compares the predictive recursion with (2), verifies every conditional martingale row, and retains exact rational values.

Total mass grows from $\sum_i a_i$ to $\sum_i a_i + cn$. Therefore this owner has no nontrivial periodic source orbit, primitive repetition clock, or source Artin–Mazur product. It supplies no arithmetic local datum, Euler factor, root number, automorphy statement, target divisor, functional equation, or Hilbert–Pólya operator. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, Route B is disabled.

References

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[2] D. Blackwell and J. B. MacQueen, *Ferguson distributions via Pólya urn schemes*, *Ann. Statist.* **1** (1973), 353–355, doi:10.1214/aos/1176342372.